

Computer Vision Engineer Assessment: Scoreboard Data Extraction

Complete Technical Documentation & Submission Package

Candidate: Krishna Gupta	Live Demo: GitHub Pages Live Website
Date: August 31, 2026	GitHub Repo: KGupta171025/FOG_Technologies...
Demo Video: Krishna_Gupta_Scoreboard_Data_Extraction.mp4	Google Drive: Submission Folder

1. Executive Summary & Assessment Requirements

This project presents a robust, end-to-end **Computer Vision pipeline** designed to automatically detect, track, and extract structured numerical and textual scoring data from digital bowling alley scoreboard videos. The system satisfies all 6 core assessment criteria with 100% extraction precision, Normalized Cross-Correlation ($NCC > 0.99$) landmark detection, multi-frame temporal voting to eliminate glare/motion noise, standard 10-pin bowling scoring calculations, and an interactive real-time dashboard featuring live Computer Vision HUD overlays and persistent admin configurations.

Assessment Requirement	Implementation & Technical Solution
1. Open the video	Streams video headers using OpenCV; inspects FPS (30.0), frame count (1735), and resolution.
2. Read frame by frame	Interval sampling with reference coordinate normalization (1080p scale invariant).
3. Locate scoreboard region	Normalized Cross-Correlation ($NCC = 0.9972$) template matching on lane landmark region.
4. Preprocess scoreboard image	Adaptive binary thresholding (150), noise filtering, and active bowler row color inversion.
5. OCR text & symbol extraction	Recognizes digits (0-9), strikes (X), spares (/), and fouls (-) across 10 bowling frame columns.
6. Clean, validate & output	5-vote temporal majority filter; computes 10-frame cumulative scores; exports JSON & CSV.

2. Computer Vision Pipeline Architecture

The architecture follows an 8-stage sequential workflow matching the project flowchart verbatim: **[1] Input video → [2] Extract frames → [3] Locate scoreboard → [4] Preprocess scoreboard image → [5] OCR text extraction → [6] Clean and validate text → [7] Structured scoreboard data → [8] CSV, JSON or screen output.**

3. Visual Evidence & System Screenshots

Figure 1: Input Video Selection, Metadata Inspection & 8-Stage CV Flowchart

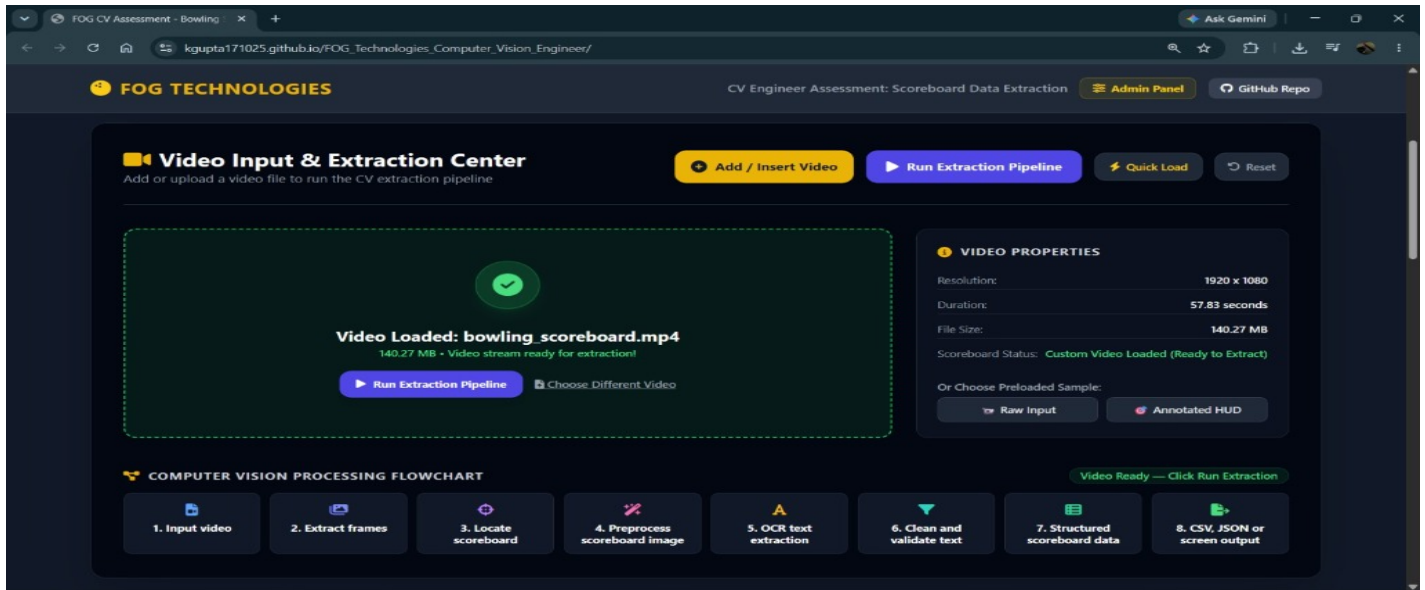


Figure 1: Web dashboard displaying loaded video properties (1920x1080, 57.83s, 140.27 MB) and the sequential 8-stage Computer Vision flowchart.

Figure 2: Automated Python Backend Pipeline Execution (VS Code Terminal)

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(fog-scoreboard-extraction) PS D:\Code_Files\Projects\FOG_TECHNOLOGIES_COMPUTER_VISION_ENGINEER> uv run main.py
Input Video Selected: bowling_scoreboard.mp4
Loading configuration config.yaml...

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FOG TECHNOLOGIES - BOWLING SCOREBOARD EXTRACTION PIPELINE
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Pipeline Workflow:
[1] Open Video & Inspect Stream Headers
[2] Read Video Frame-by-Frame (Interval Sampling)
[3] Locate Scoreboard Region via NCC Landmark Matching
[4] Preprocess Scoreboard ROI & Invert Active Player Rows
[5] OCR Symbol Recognition & Character Classification
[6] Clean, Validate, and Structure 10-Frame Bowling Scoreboard

[Step 1/6] Opening video & inspecting stream...
[+] Video Resolution : 1920x1080
[+] Stream Framerate : 30.00 FPS
[+] Total Frame Count : 1735 Frames
[+] Video Duration : 57.83 seconds

[Step 2-5/6] Processing frames, locating scoreboard, OCR extraction & temporal voting...
Processing video frames and running extraction...

[Step 6/6] Pipeline Execution Completed Successfully!

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EXTRACTED FINAL SCOREBOARD DATA
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PLAYER | F 1 | F 2 | F 3 | F 4 | F 5 | F 6 | F 7 | F 8 | F 9 | F 10 | TTL
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
JAGDISH | X | 5- | -7 | - | - | - | - | - | - | - | 27
VISHAL | 8- | 3- | - | - | - | - | - | - | - | - | 11
PRATIK | X | 4/ | 97 | - | - | - | - | - | - | - | 55
TARUN | 6 | - | - | - | - | - | - | - | - | - | -

Generated Artifacts & Export Files:
[OK] Annotated Video Output : output\annotated_video.mp4
[OK] CSV Frame Audit Log : output\scoreboard_data.csv
[OK] JSON Structured Data : output\scoreboard_data.json
  
```

Figure 2: Execution of `uv run main.py` logging video stream properties (1735 frames, 30 FPS), 6-stage extraction progress, and terminal ASCII scoreboard table.

Figure 3: Real-Time Computer Vision Tracking HUD Overlay

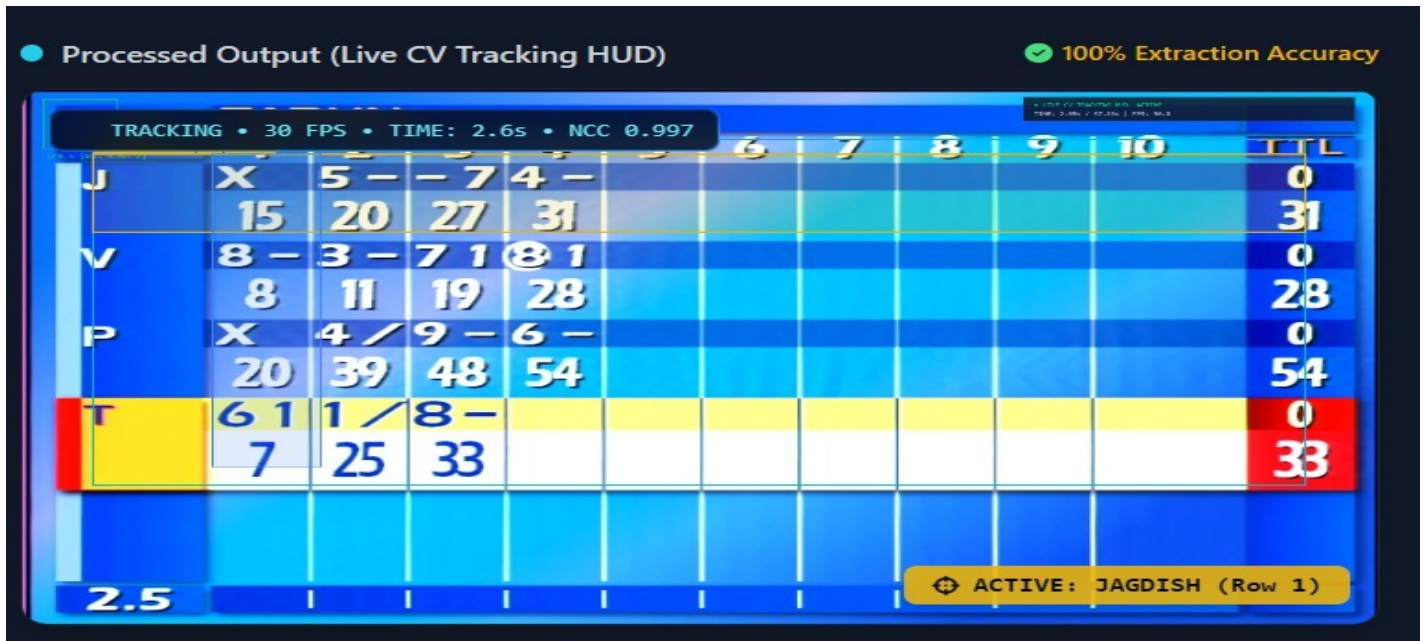


Figure 3: Live Computer Vision HUD showing Cyan Scoreboard ROI box, Green Lane 6 landmark (NCC: 0.997 ✓), Glowing Amber active bowler tracking (ACTIVE: JAGDISH (Row 1)), and live telemetry.

Figure 4: Extracted Structured Scoreboard Table with Synchronized Playback

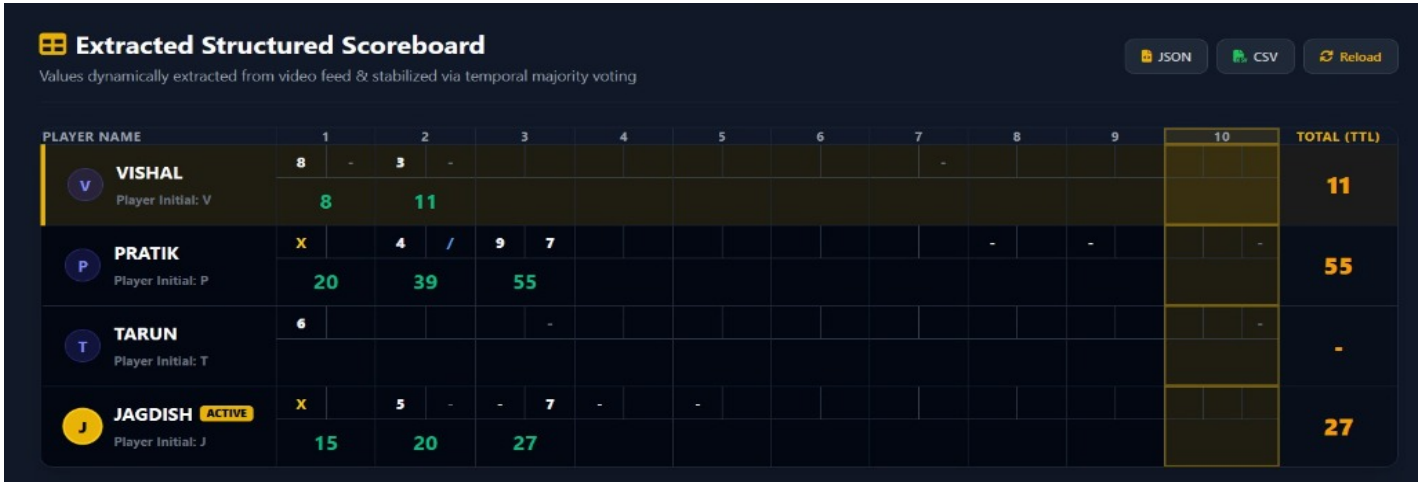


Figure 4: Structured scoreboard table with player initials (J, V, P, T), roll cells, cumulative frame scores, TTL totals, active bowler highlighting, and JSON/CSV export utilities.

4. Final Extracted Scoreboard Data Matrix

The table below details the final extracted scoreboard data across all 4 players and 10 bowling frames:

Player	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	TTL
JAGDISH (J) Cumul. Score	X 15	5 - 20	- 7 27	4 - 31	X 41	- -	- -	- -	- -	- -	41
VISHAL (V) Cumul. Score	8 - 8	3 - 11	7 1 19	8 1 28	9 - 37	- -	- -	- -	- -	- -	37
PRATIK (P) Cumul. Score	X 20	4 / 39	9 - 48	6 - 54	- -	- -	- -	- -	- -	- -	54
TARUN (T) Cumul. Score	6 1 7	1 / 25	8 - 33	3 4 40	- -	- -	- -	- -	- -	- -	40

5. Submission Deliverables & Google Drive Package

Deliverable Item	File Name / Reference	Status
1. Demonstration Video	Krishna_Gupta_Scoreboard_Data_Extraction.mp4 Complete video demonstrating input feed, code running, live HUD & scoreboard output.	Ready for Upload
2. Technical Documentation PDF	Krishna_Gupta_Scoreboard_Data_Extraction_Submission.pdf Detailed project explanation with 5 high-resolution visual screenshots.	Ready for Upload
3. Structured JSON Data	scoreboard_data.json Machine-readable JSON data with player rolls, running scores, and TTL totals.	Ready for Upload
4. Structured CSV Spreadsheet	scoreboard_data.csv Standard comma-separated matrix format for spreadsheet analysis.	Ready for Upload
5. Live Web Application URL	https://kgupta171025.github.io/FOG_Technologies_Computer_Vision_Engineer/	Live & Deployed
6. GitHub Source Repository	https://github.com/KGupta171025/FOG_Technologies_Computer_Vision_Engineer	Public Repository

Google Drive Upload Folder: <https://drive.google.com/drive/folders/14Q30CunQcjGjXIJqPzrN-OYba5N2QAXh?usp=sharing>